INNOVATIVE METALS COMPANY, INC.

IRF Purlin Retrofit Analysis

Project Name:	After analyses
Project Reference:	—
Project Location:	—
Client:	—
Design Report Date:	10/30/2026
Revision No.:	0

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1. Project

Client Name	—
Project Name	After analyses
Project Location	—
Project Reference	—
Design Method	ASD, AISI S100-16
Design Report Date	10/30/2026
Time	11:14:07
Revision No.	0

This report gives the allowable roof pressure of the existing purlin line and of the same line after the IRF retrofit, for gravity and uplift loading. For each case it reports the limit state that governs, the location along the line where it governs, and the maximum vertical deflection at the allowable pressure.

2. Analysis Methods

1. **Analysis.** The purlin line is solved as a continuous thin-walled beam under a uniform roof pressure, with the roof deck acting as continuous lateral and rotational bracing on the top flange (both decks in the retrofit). The free bottom flange is modelled separately as a beam-column. The analysis is second order, so lateral-torsional buckling is included in the demands.
2. **Purlin line segments.** The line is analysed as one continuous member. Within the lap length on each side of an interior support the two overlapping purlins act together, so that segment is given a section with the combined thickness of the two purlins; every strength is calculated separately for each segment, and the lower of two adjacent segments' strengths is used where they meet.
3. **Strengths.** Section strengths follow AISI S100-16 (ASD): flexure by the Direct Strength Method with local, distortional and net-section (punch-out) effects, shear, torsion, and web crippling at the supports. In the retrofit the purlin and the IntelliFrame act as one section.
4. **Allowable pressure.** The pressure is increased until the governing demand-to-capacity ratio along the line reaches 1.0. The limit states checked are flexural + torsional interaction (H4.2), biaxial bending (H1.2), flexure + shear (H2.1), distortional buckling (F4) and web crippling (G5). The deflection is reported at that pressure.

3. Existing Roof

3.1. Build

Table 1: Existing purlin line

Purlin	Z8x2.5 054
Spans	18 / 22 / 18 ft (3 spans, 58.0 ft total)
Purlin lap at each interior support	2 ft each side of the support
Purlin spacing	5 ft
Roof slope	1.0:12
Existing roof deck	SSR Ultra-Dek 18 in. 24 ga
Purlin-to-frame connection	Clip-mounted
Frame flange width	6.0 in.

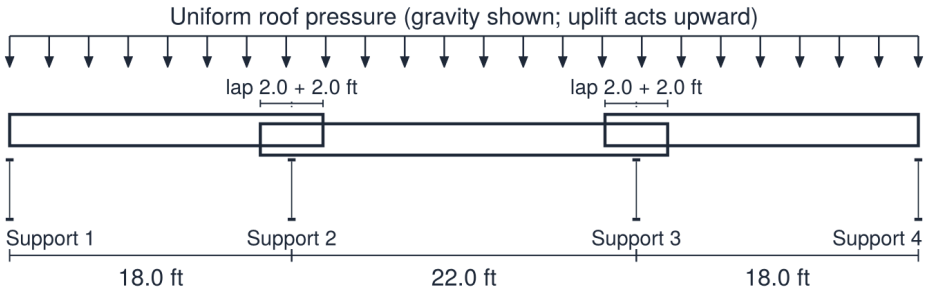


Figure 1: Span configuration: purlins lapped over the interior frame supports, span lengths in ft, frame flange width 6.0 in. (horizontal to scale, vertical exaggerated 3×)

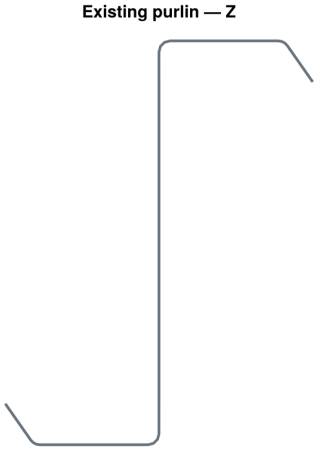


Figure 2: Existing purlin, Z8x2.5 054

Table 2: Segments along the purlin line, left to right

Segment	Length	Section	t (in.)	Type
1	16.0 ft	Z8x2.5 054	0.054	Span
2	2.0 ft	Z8x2.5 054 + Z8x2.5 054 (lap)	0.108	Lap (two purlins)
3	2.0 ft	Z8x2.5 054 + Z8x2.5 054 (lap)	0.108	Lap (two purlins)
4	18.0 ft	Z8x2.5 054	0.054	Span
5	2.0 ft	Z8x2.5 054 + Z8x2.5 054 (lap)	0.108	Lap (two purlins)
6	2.0 ft	Z8x2.5 054 + Z8x2.5 054 (lap)	0.108	Lap (two purlins)
7	16.0 ft	Z8x2.5 054	0.054	Span

3.2. Gravity

Table 3: Existing purlin line, gravity

Allowable roof pressure	**14.8 psf** (74 plf on the purlin)
Governing limit state	Flexure, biaxial bending (AISI S100 H1.2)
Failure location	6.2 ft from the left end, in span 1, 6.2 ft from support 1
Maximum vertical deflection	0.47 in. (L/560) in span 2, 29.0 ft from the left end

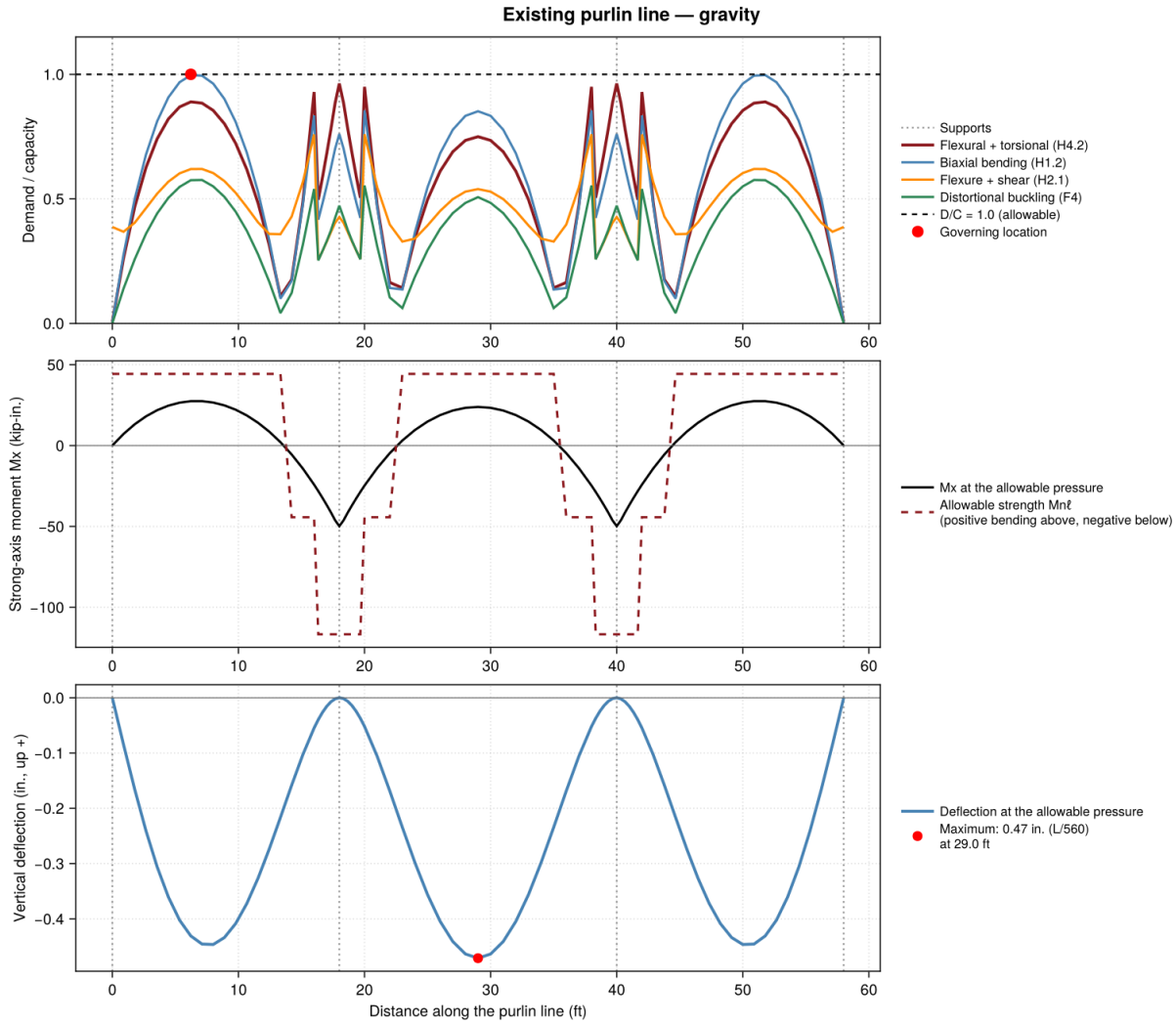


Figure 3: Existing purlin line, gravity – demand-to-capacity ratios at the allowable pressure (red circle: governing location), strong-axis moment against the allowable strength, and vertical deflection

3.3. Uplift

Table 4: Existing purlin line, uplift

Allowable roof pressure	**11.4 psf** (57 plf on the purlin)
Governing limit state	Flexural + torsional interaction (AISI S100 H4.2)
Failure location	6.2 ft from the left end, in span 1, 6.2 ft from support 1
Maximum vertical deflection	0.40 in. (L/652) in span 2, 29.0 ft from the left end

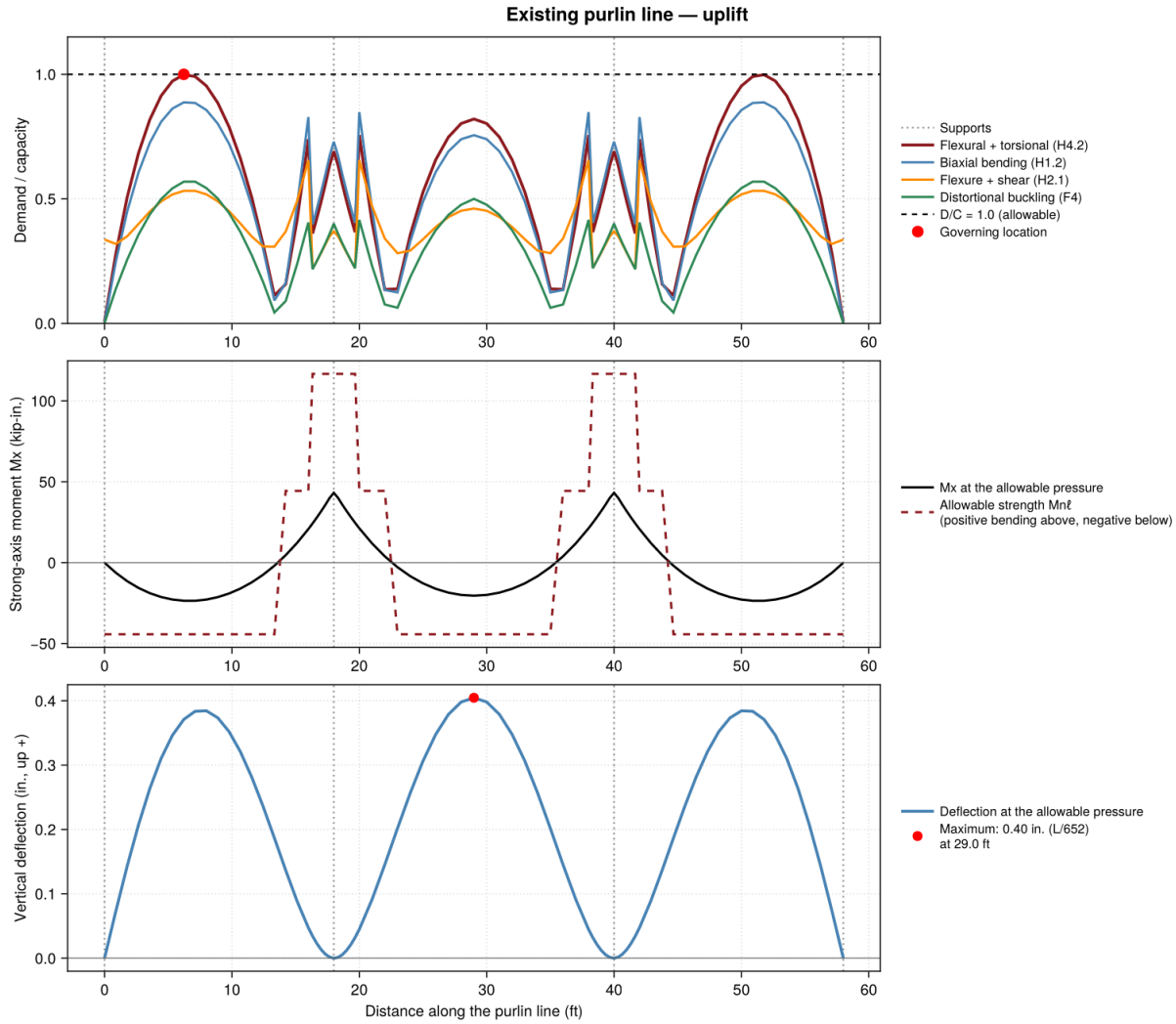


Figure 4: Existing purlin line, uplift – demand-to-capacity ratios at the allowable pressure (red circle: governing location), strong-axis moment against the allowable strength, and vertical deflection

4. IntelliFrame RF Retrofit

4.1. Build

Table 5: IntelliFrame and new roof

IntelliFrame section	2x4.5x2.5 14g
Thickness t	0.071 in.
Bottom flange / web / top flange / lip	2.00 / 4.50 / 2.50 / 0.50 in.
Punch-out (height × length)	3.000 × 2.000 in.
Steel	F_y = 55 ksi, E = 29500 ksi
New roof deck	SSR Ultra-Dek 18 in. 24 ga
Attachment	IntelliFrame screwed through the existing deck to the purlin top flange; new deck fastened to the IntelliFrame top flange

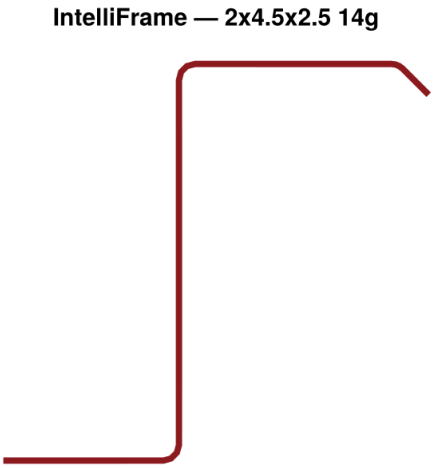


Figure 5: IntelliFrame 2x4.5x2.5 14g

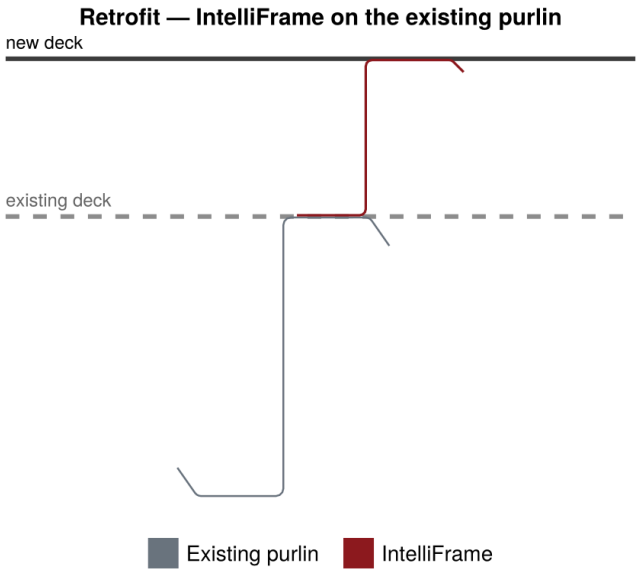


Figure 6: Retrofit assembly: IntelliFrame on the existing purlin, between the existing and the new deck

Table 6: IntelliFrame local buckling moment, positive bending (kip-in.) – governing value per segment

Segment	Away from punch-out			At punch-out		
	combined	bare (cFSM)	**governing**	combined	bare (cFSM)	**governing**
1	306	471	**306**	297	423	**297**
2	353	584	**353**	359	523	**359**
3	353	584	**353**	359	523	**359**
4	306	471	**306**	297	423	**297**
5	353	584	**353**	359	523	**359**
6	353	584	**353**	359	523	**359**
7	306	471	**306**	297	423	**297**

Bare IntelliFrame: $M_{cr\ell}$ = 112 kip-in. away from the punch-out and 23.1 kip-in. at the punch-out, about its own centroid, before conversion to the combined section.

4.2. Gravity

Table 7: IntelliFrame RF retrofit, gravity

Allowable roof pressure	**28.8 psf** (144 plf on the purlin)
Governing limit state	Flexural + torsional interaction (AISI S100 H4.2)
Failure location	18.0 ft from the left end, at interior support 2
Maximum vertical deflection	0.21 in. (L/1270) in span 2, 29.0 ft from the left end

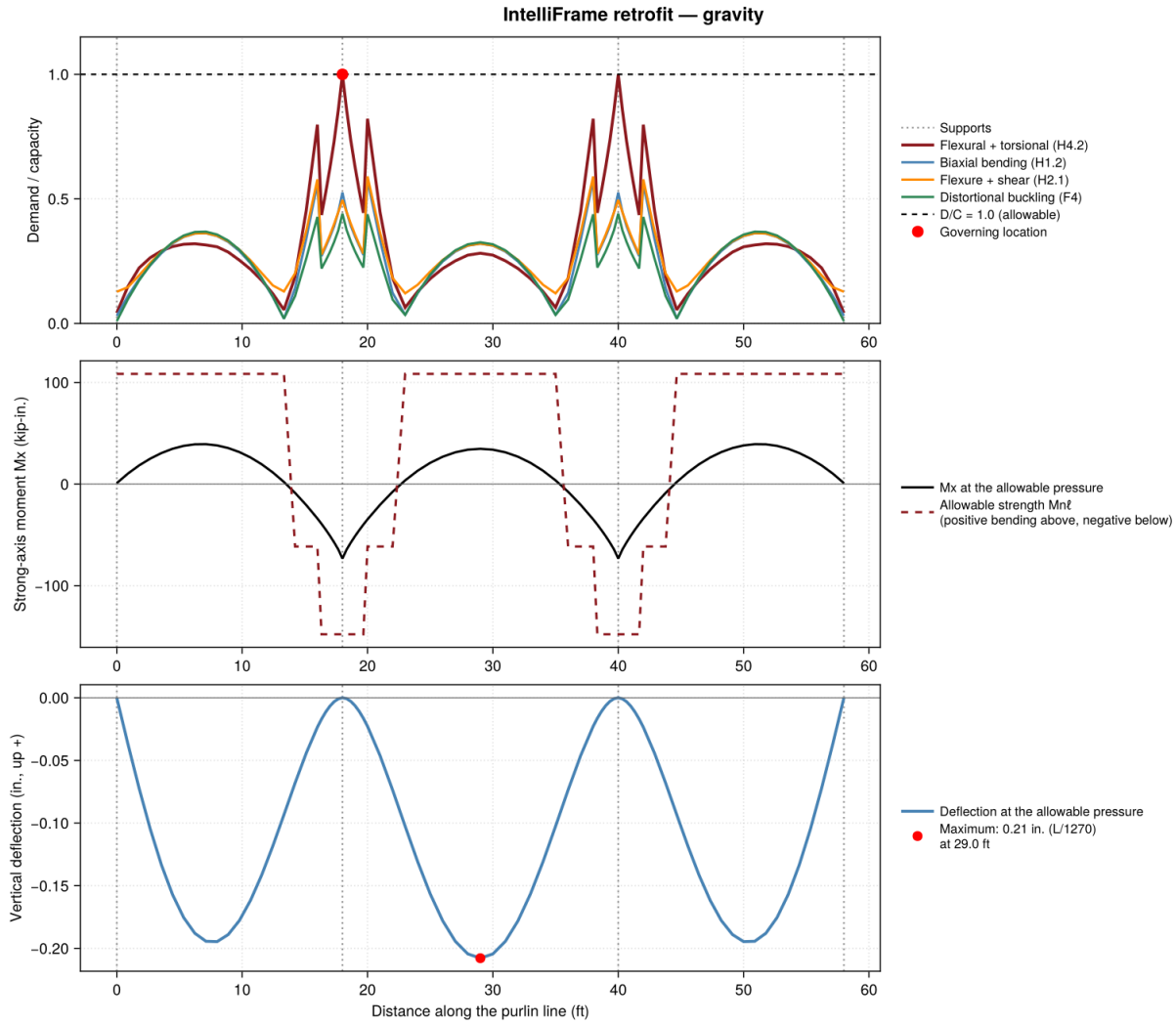


Figure 7: IntelliFrame RF retrofit, gravity – demand-to-capacity ratios at the allowable pressure (red circle: governing location), strong-axis moment against the allowable strength, and vertical deflection

4.3. Uplift

Table 8: IntelliFrame RF retrofit, uplift

Allowable roof pressure	**29.2 psf** (146 plf on the purlin)
Governing limit state	Flexural + torsional interaction (AISI S100 H4.2)
Failure location	6.2 ft from the left end, in span 1, 6.2 ft from support 1
Maximum vertical deflection	0.21 in. (L/1250) in span 2, 29.0 ft from the left end

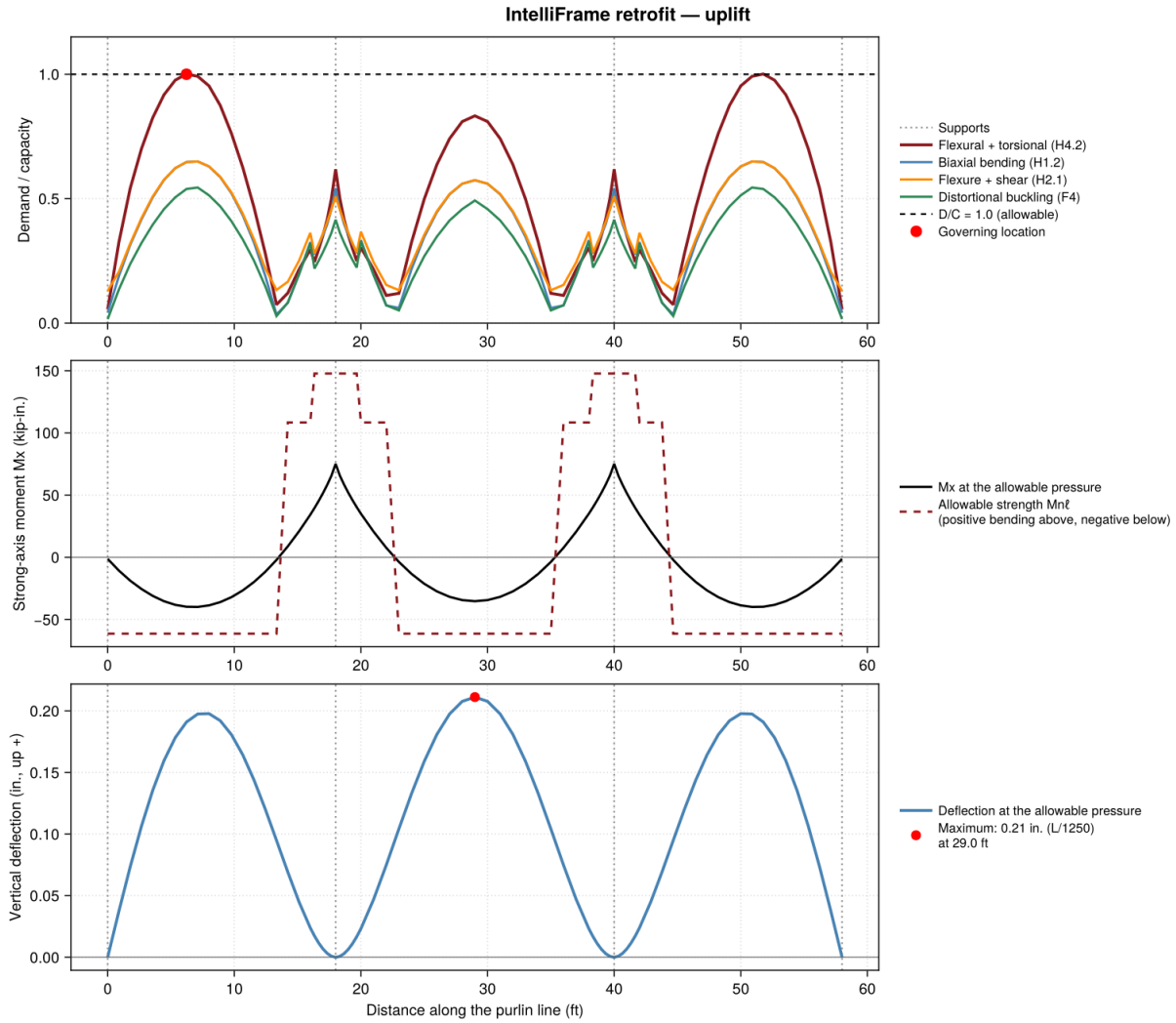


Figure 8: IntelliFrame RF retrofit, uplift – demand-to-capacity ratios at the allowable pressure (red circle: governing location), strong-axis moment against the allowable strength, and vertical deflection

4.4. Allowable flexural strengths

Table 9: Allowable strong-axis flexural strength $M_{n\ell}$ (kip-in.) per segment, existing vs. retrofit

Segment	Existing purlin		IntelliFrame RF retrofit	
	positive (+)	negative (−)	positive (+)	negative (−)
1	44.3	44.3	108	61.4
2	117	117	148	148
3	117	117	148	148
4	44.3	44.3	108	61.4
5	117	117	148	148
6	117	117	148	148
7	44.3	44.3	108	61.4

5. Summary

Table 10: Allowable roof pressure and governing limit state

	Gravity		Uplift	
	Existing	IRF retrofit	Existing	IRF retrofit
Allowable pressure	**14.8 psf**	**28.8 psf**	**11.4 psf**	**29.2 psf**
Improvement	+95 %		+155 %	
Governing limit state	Flexure, biaxial bending (AISI S100 H1.2)	Flexural + torsional interaction (AISI S100 H4.2)	Flexural + torsional interaction (AISI S100 H4.2)	Flexural + torsional interaction (AISI S100 H4.2)
Failure location	In span 1, 6.2 ft from support 1	At interior support 2	In span 1, 6.2 ft from support 1	In span 1, 6.2 ft from support 1
Max. deflection	0.47 in. (L/560)	0.21 in. (L/1270)	0.40 in. (L/652)	0.21 in. (L/1250)

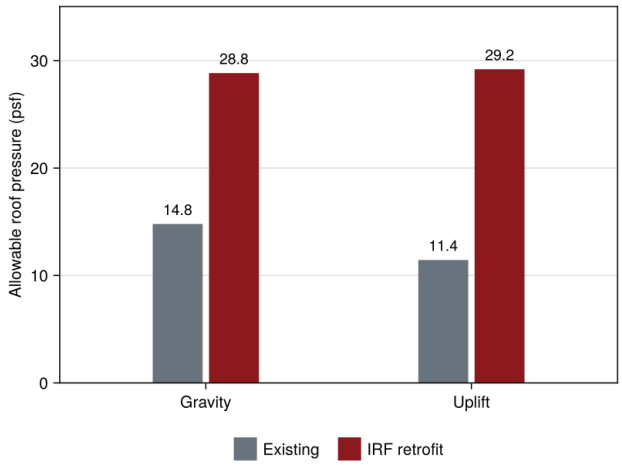


Figure 9: Allowable roof pressure, existing vs. IRF retrofit

The IRF retrofit raises the allowable gravity pressure of the purlin line from 14.8 psf to 28.8 psf and the allowable uplift pressure from 11.4 psf to 29.2 psf. Deflections are reported at the allowable pressure; the IRF stiffens the line, so the retrofitted deflection at the higher pressure is still smaller than that of the existing line.